

Credit Card Approval Prediction

Project Description:

The Credit Card Approval Prediction System is a Machine Learning-based web application that automates the process of evaluating credit card applications. Financial institutions receive thousands of applications every day, making manual verification slow, expensive, and susceptible to human errors. This project aims to simplify and accelerate the approval process by predicting whether a credit card application is likely to be approved or rejected based on the applicant's financial and demographic information.

The system utilizes four supervised machine learning algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost. Historical credit card applicant data is preprocessed, cleaned, and used to train these models. Their performances are compared using evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The best-performing model is saved using Pickle and integrated into a Flask web application for real-time predictions.

To make the application scalable and cloud-accessible, the trained model is deployed using IBM Watson Machine Learning. Users can access the prediction system through an intuitive web interface where they enter applicant details and instantly receive an approval prediction.

Scenarios

Scenario 1: Automated Credit Card Application Screening

A bank employee enters an applicant's details such as annual income, employment status, loan amount, credit history, and existing debts into the web application. The machine learning model instantly predicts whether the application should be approved or rejected, reducing manual workload and improving decision-making speed.

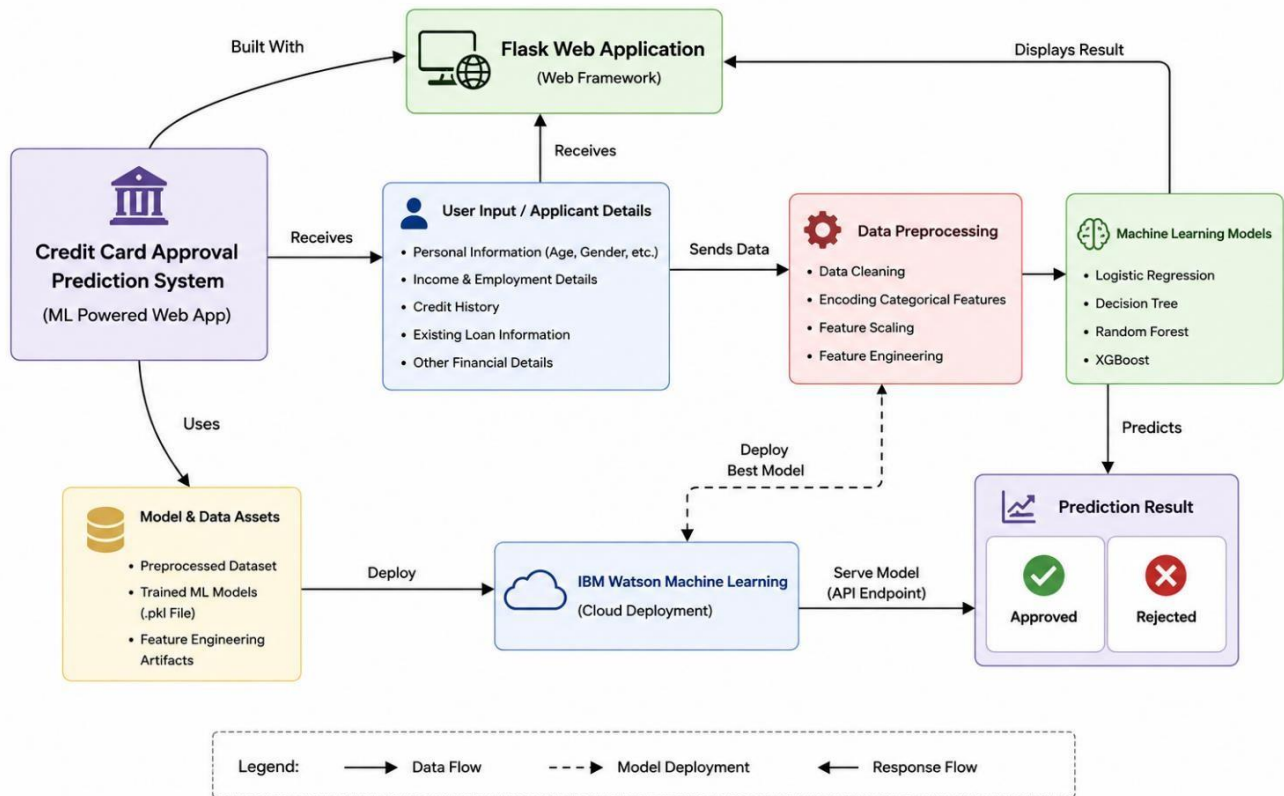
Scenario 2: High-Risk Applicant Identification

A compliance officer screens applicants with previous loan defaults and poor repayment records. The system identifies applicants with high financial risk and recommends rejection, helping banks reduce loan defaults and financial losses.

Scenario 3: Customer Self-Service Eligibility Check

A customer visits the web portal before applying for a credit card. After entering financial information such as income, employment status, and credit history, the system predicts the likelihood of approval, allowing customers to understand their eligibility before submitting an official application.

Technical Architecture:



Description: The Credit Card Approval Prediction System is built using the Flask web framework, which provides a user-friendly interface for entering applicant details. The application preprocesses the input data and sends it to trained machine learning models, including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model predicts whether the credit card application will be approved or rejected based on the applicant's financial profile. The trained model is deployed using IBM Watson Machine Learning to enable secure and scalable cloud-based predictions. Finally, the prediction result is displayed instantly to the user through the Flask web application.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

sPre-requisites:

- **Python Programming Proficiency:** Python Documentation
- **Machine Learning Fundamentals:** Scikit-learn Documentation
- **Flask Framework Knowledge:** Flask Documentation
- **Pandas and NumPy for Data Processing:** Pandas & NumPy Documentation
- **XGBoost Library Familiarity:** XGBoost Documentation
- **HTML, CSS, and JavaScript Skills:** W3Schools HTML/CSS/JavaScript Tutorials
- **IBM Watson Machine Learning Knowledge:** IBM Cloud & Watson Machine Learning Documentation
- **Model Serialization using Pickle:** Python Pickle Documentation
- **Version Control with Git:** Git Documentation
- **Development Environment Setup:** Flask Installation Guide

Dataset Requirements

- Historical Credit Card Approval Dataset
- Applicant Financial Information
- Credit History Records
- Income and Employment Details

Python Libraries

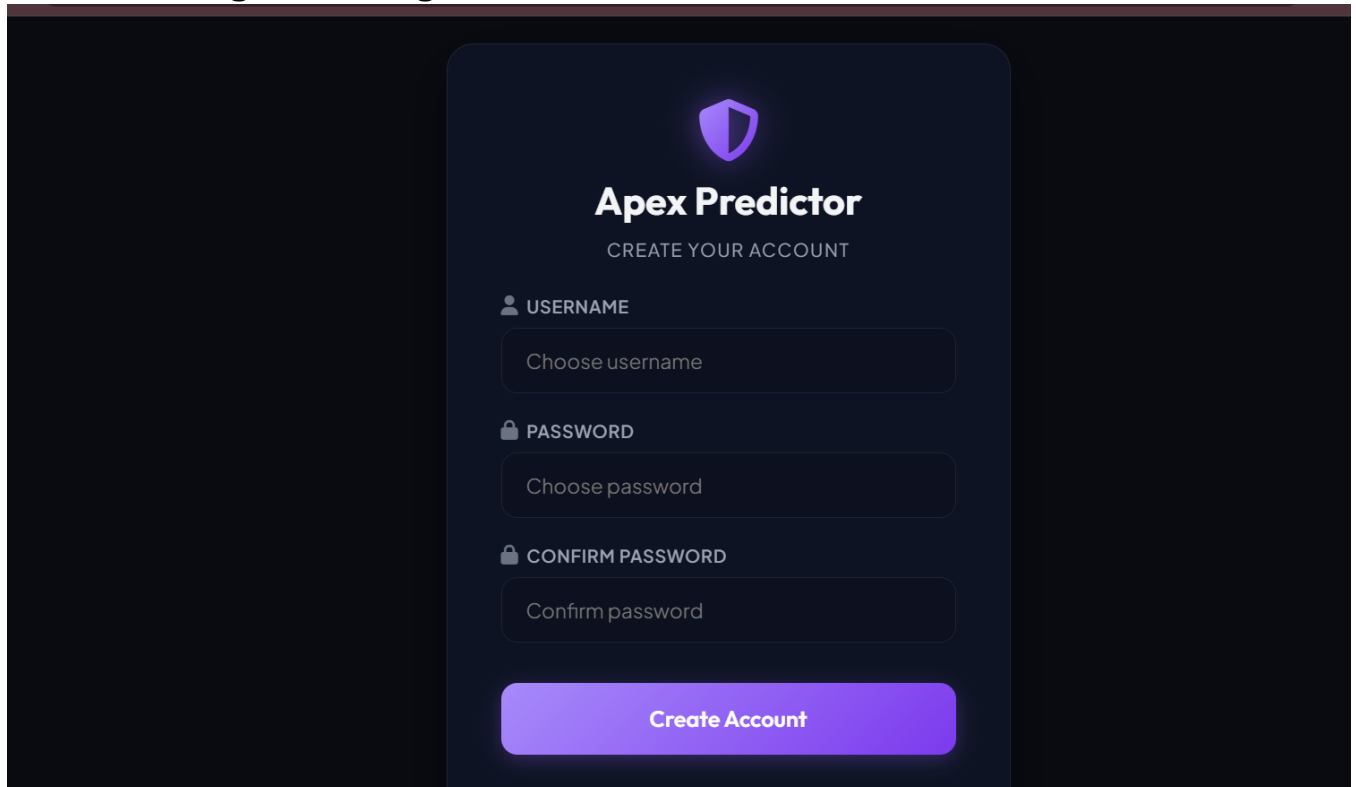
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Flask
- Joblib/Pickle
- Matplotlib

Technical Knowledge

- Python Programming
- Machine Learning Basics
- Data Preprocessing Techniques
- Classification Algorithms
- Flask Web Development
- HTML, CSS, and JavaScript
- REST API Basics
- Version Control using Git

Project Workflow:

Milestone 1: Registration Page



The registration page features a dark blue background with a central white rounded rectangle. At the top of the rectangle is a shield icon. Below the icon, the text "Apex Predictor" is displayed in a large, bold font, followed by "CREATE YOUR ACCOUNT" in a smaller font. The form includes three input fields: "USERNAME" with a placeholder "Choose username", "PASSWORD" with a placeholder "Choose password", and "CONFIRM PASSWORD" with a placeholder "Confirm password". Each field is preceded by a small icon (person for username, lock for password). A large blue button labeled "Create Account" is positioned at the bottom of the form.

Apex Predictor
CREATE YOUR ACCOUNT

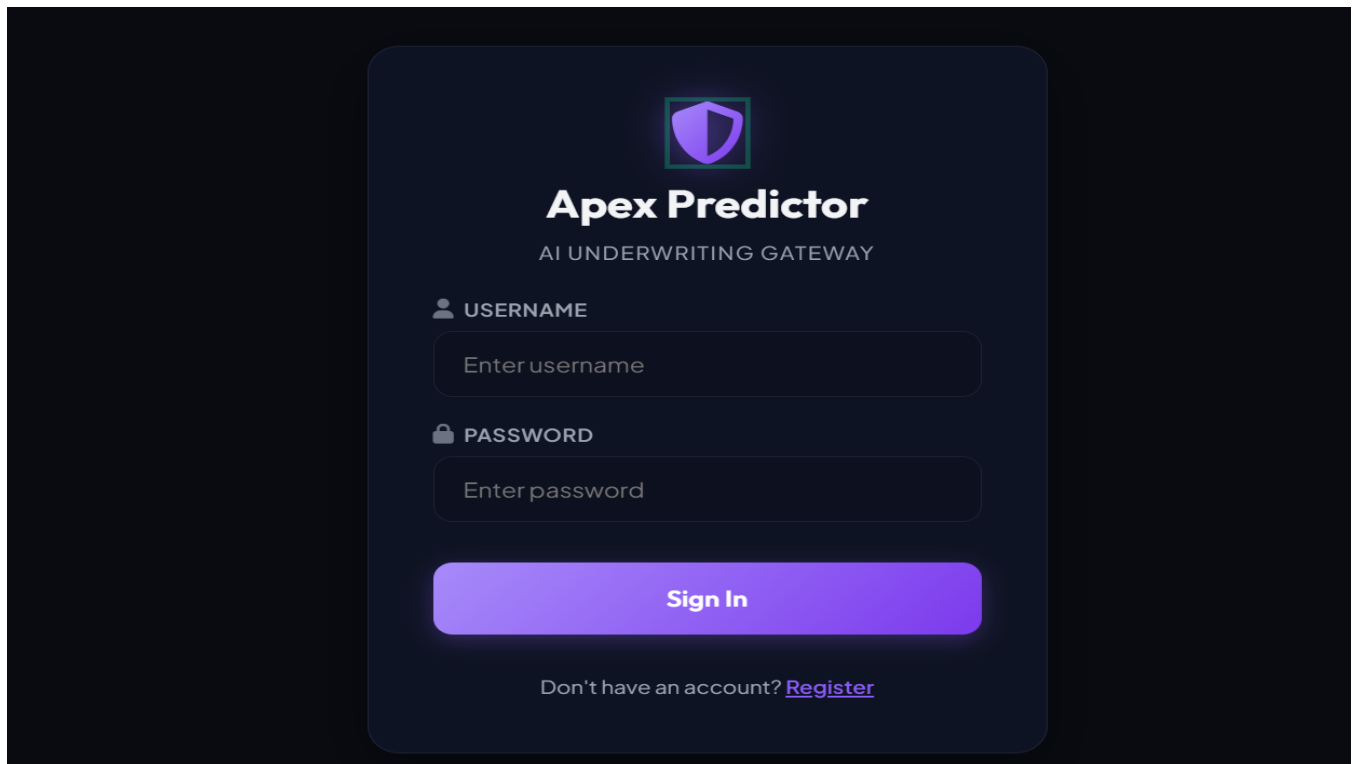
USERNAME
Choose username

PASSWORD
Choose password

CONFIRM PASSWORD
Confirm password

Create Account

Milestone 2: Login Page



The login page features a dark blue background with a central white rounded rectangle. At the top of the rectangle is a shield icon. Below the icon, the text "Apex Predictor" is displayed in a large, bold font, followed by "AI UNDERWRITING GATEWAY" in a smaller font. The form includes two input fields: "USERNAME" with a placeholder "Enter username" and "PASSWORD" with a placeholder "Enter password". Each field is preceded by a small icon (person for username, lock for password). A large blue button labeled "Sign In" is positioned at the bottom of the form. Below the button, there is a link that says "Don't have an account? [Register](#)".

Apex Predictor
AI UNDERWRITING GATEWAY

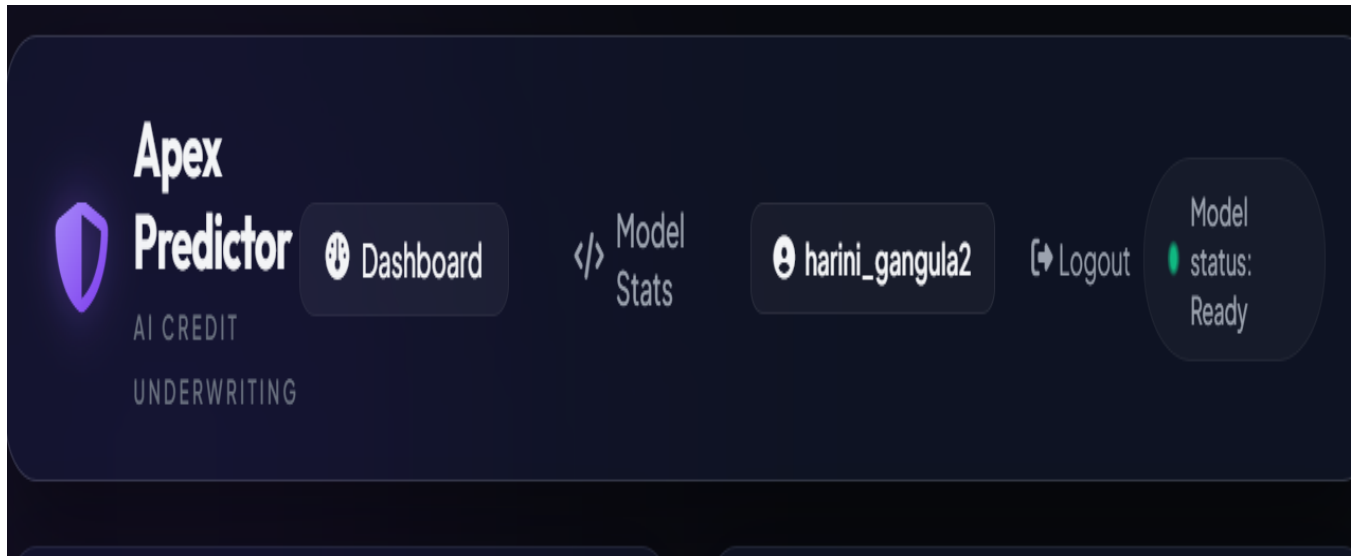
USERNAME
Enter username

PASSWORD
Enter password

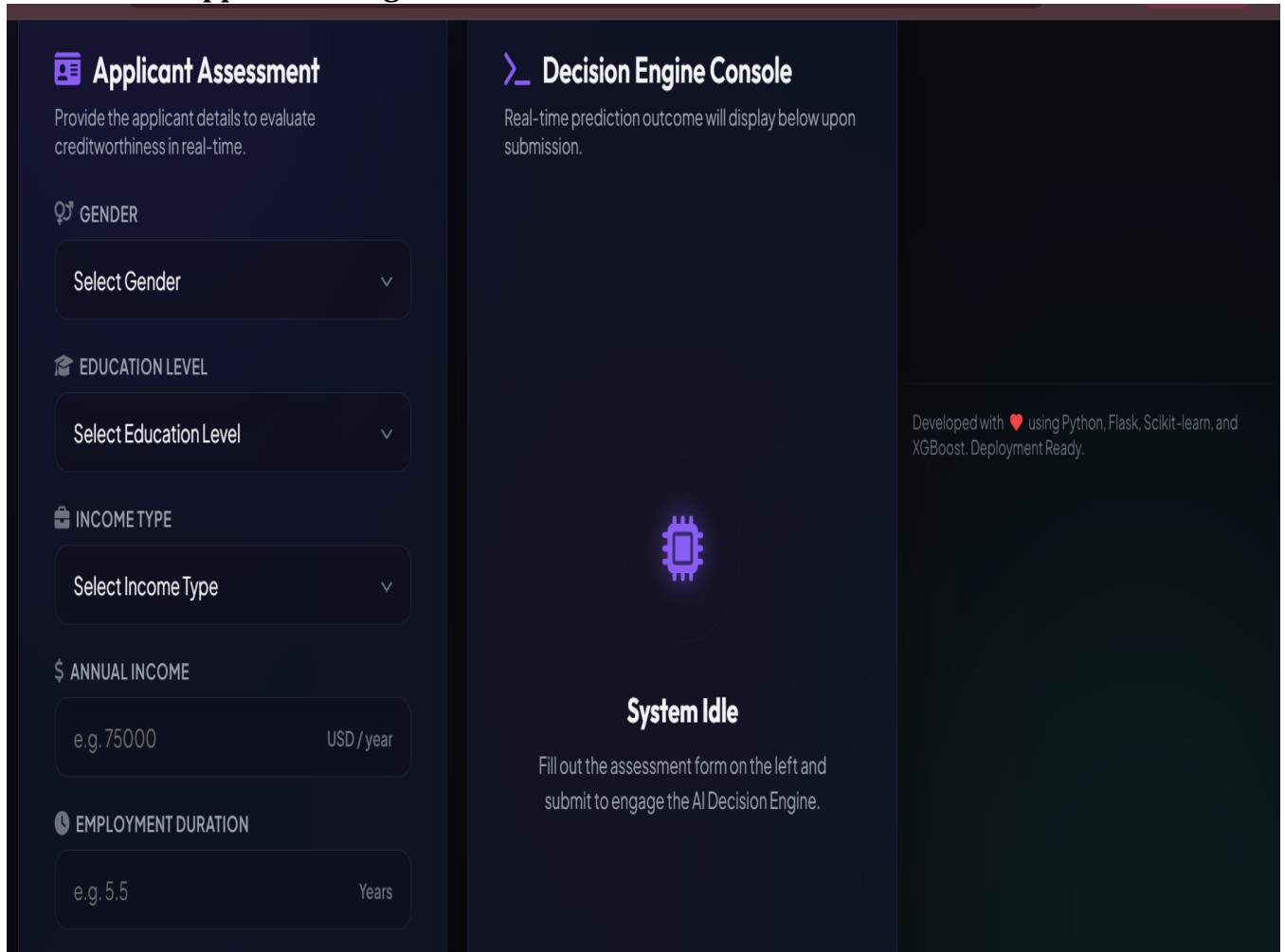
Sign In

Don't have an account? [Register](#)

Milestone 3 : User Profile



Milestone 4: Application Page



Milestone 5: Evaluation Page(Approved)

Applicant Assessment

Provide the applicant details to evaluate creditworthiness in real-time.

GENDER

Female

EDUCATION LEVEL

Master's Degree

INCOME TYPE

Business Owner

ANNUAL INCOME

85000 USD / year

EMPLOYMENT DURATION

6.5 Years

Decision Engine Console

Real-time prediction outcome will display below upon submission.

APPLICATION APPROVED

97.4%
CONFIDENCE

Decision	Approved
Active ML Model	Random Forest
Risk Threshold	< 50% Rejection Probability

Applicant meets creditworthiness criteria. High propensity for repayment. Approval recommended.

Milestone 6: Evaluation Page(Rejected)

Applicant Assessment

Provide the applicant details to evaluate creditworthiness in real-time.

GENDER

Male

EDUCATION LEVEL

High School

INCOME TYPE

Business Owner

ANNUAL INCOME

40000 USD / year

EMPLOYMENT DURATION

5.5 Years

Decision Engine Console

Real-time prediction outcome will display below upon submission.

APPLICATION REJECTED

91.0%
CONFIDENCE

Decision	Rejected
Active ML Model	Random Forest
Risk Threshold	> 50% Rejection Probability

Applicant falls below minimum score threshold. Risk index elevated. Reject recommended.

Conclusion:

The Credit Card Approval Prediction System successfully automates the process of evaluating credit card applications using machine learning techniques. By comparing multiple classification algorithms—Logistic Regression, Decision Tree, Random Forest, and XGBoost—the project identifies the best-performing model to accurately predict whether an application should be approved or rejected. The trained model is integrated into a Flask web application, providing users with instant and reliable predictions through a simple interface. Furthermore, deployment using IBM Watson Machine Learning enables secure, scalable, and cloud-based access to the prediction service. This project helps financial institutions reduce manual effort, improve decision-making accuracy, and speed up the credit approval process, while offering a strong foundation for future enhancements such as explainable AI, fraud detection, and real-time banking system integration.